

Pointing method in Boltzmann Sampler

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1 Introduction

1.1 Background

Several different methods have been used for the random generation of combinatorial structures. Ad hoc algorithms [1] using known bijections offer advantageous complexity but little to none portability. Markov chains obtain the wanted distribution on a combinatorial class by exploring the class using little steps. This method still lack movability. A more general approach is the recursive method [3] which is based on counting elements. A major drawback is the high complexity appearing in those algorithm. All three methods are used to obtain determined sized objects. By relaxing this condition and allowing a window of a few percent on the size, the Boltzmann Samplers introduced in 2004 offer high portability with a linear complexity in most studied cases [4].

1.2 Plan

The first section gives a general overview for the random generation of combinatorial structures. The second section is separated into three parts. First we introduce the analytic combinatorics. Then, we present Boltzmann method and Sampler. Finally, we offer very simple results on this sampler. This will give a solid theory to this paper and help build an intuition on Boltzmann Samplers. The third and final section introduces the pointing method with justifications on the importance of this method.

There are also two appendix. The first introduce more advanced combinatorial operations, those operations are too complex to be studied in such a short paper and will only be presented briefly. The second presents a particular type of Boltzmann Samplers: the singular samplers. Again, those samplers will only be presented briefly, this time with an example.

2 Combinatorial theory and Boltzmann Samplers

2.1 Analytic Combinatorics

This section introduce the theory behind combinatorial class and analytic combinatoric which core idea is to associate to a combinatorial class a power series.

First we define the class of combinatorial structures in a similar way to [3]:

Definition 1 (Combinatorial Structures) *A class of combinatorial structures is a pair $\mathcal{C} = \langle \mathcal{C}, |\cdot|_{\mathcal{C}} \rangle$, where $\mathcal{C}$ is a finite or denumerable set and $|\cdot|_{\mathcal{C}}$ is an application from $\mathcal{C}$ to $\mathbb{N}$.*

For $c \in \mathcal{C}$, $|c|_{\mathcal{C}}$ is the size of c . There must always be a finite number of elements of each size in $\mathcal{C}$.

The counting sequence $(C_n)_{n \in \mathbb{N}}$ associated with $\mathcal{C}$ is the number of elements of each size in $\mathcal{C}$. I.e. $\forall n \in \mathbb{N}, C_n = |\{c \in \mathcal{C} | |c|_{\mathcal{C}} = n\}|$.

Later on, we will indistinctively use $\mathcal{C}$ or $\langle \mathcal{C}, |\cdot| \rangle$. We will sometimes simplify the notation for the size: $|\cdot|$. When different class of combinatorial structures will be involved context can be used to differentiate their different size functions.

Definition 2 (Isomorphism) *Two combinatorial class are said to be isomorphic if they have the same number of elements of each size.*

We now introduce the two most frequently used and simple class of combinatorial structures. They are often used as base class to build more complex ones.

Definition 3 (Epsilon and atom) *The epsilon class $\epsilon = \langle \mathcal{E}, |\cdot|_{\epsilon} \rangle$ is the class composed of only one element of size zero: $\mathcal{E} = \{e\}$ and $|e|_{\epsilon} = 0$.*

Very similar, the atom class is comprised of exactly one element of size one: $\mathcal{Z} = \{z\}$, $|\cdot|_{\mathcal{Z}}$ with $|z|_{\mathcal{Z}} = 1$.

We now define a number of operations on those objects:

Definition 4 (Product) *Given two class of combinatorial structures, $\mathcal{A} = \langle \mathcal{A}, |\cdot|_{\mathcal{A}} \rangle$, $\mathcal{B} = \langle \mathcal{B}, |\cdot|_{\mathcal{B}} \rangle$, their product noted $\mathcal{A} \times \mathcal{B}$ is the class $\langle \mathcal{A} \times \mathcal{B}, |\cdot|_{\mathcal{A} \times \mathcal{B}} \rangle$ such that $\forall (a, b) \in \mathcal{A} \times \mathcal{B}, |(a, b)|_{\mathcal{A} \times \mathcal{B}} = |a|_{\mathcal{A}} + |b|_{\mathcal{B}}$.*

As mentionned earlier, we can simplify the notations for the size function: $|\cdot|$ and referring as a class by its set: $\mathcal{C}$.

Definition 5 (Union) *Given two class of combinatorial structures, $\mathcal{A}$, $\mathcal{B}$, we define their union $\mathcal{A} + \mathcal{B}$ as follows. Let μ and μ' such that $\mu \neq \mu'$ and $|\mu| = |\mu'| = 0$, then $\mathcal{A} + \mathcal{B} = (\{\mu\} \times \mathcal{A}) \cup (\{\mu'\} \times \mathcal{B})$. This is equivalent to changing elements of $\mathcal{B}$ such that $\mathcal{A}$ and $\mathcal{B}$ are disjoint.*

This definition can be extended to the union of a finite or denumerable number of class of combinatorial structures as long as only a finite number of elements of size n appear in the union. For instance:

Definition 6 (Sequence) *The sequence class noted $\mathcal{C} = \text{Seq}(\mathcal{A})$ is defined as $\mathcal{C} = \mathcal{E} + \mathcal{A} + \mathcal{A} \times \mathcal{A} + \mathcal{A} \times \mathcal{A} \times \mathcal{A} + \dots$. To ensure that $\mathcal{C}$ is properly defined, $\mathcal{A}$ must not contain any element of size zero. Otherwise, there would be an infinite number of elements of same size.*

Using only products, unions, sequences and finite sets to build class offer a very limited range of possibilities close to the expression level of regular languages. We can use recursive definition to obtain a range closer to context-free grammar.

Definition 7 (specification) *A specification of a class of combinatorial structures $\mathcal{C}$ is a set of (possibly recursive) equations using only operations described above and atoms (defined as class with exactly one element of size one) such that $\mathcal{C}$ is the unique solution to the primary equation (up to an isomorphism).*

$\mathcal{C}$ is said to be specifiable if it admits a specification of finite size.

This definition means that for any specifiable class $\mathcal{C}$, an element $c \in \mathcal{C}$ is composed of n atoms (in this context an atom is an element of size 1). Due to above definitions, each of these atoms is distinct from the over.

An alternate definition for sequences is by giving it a specification: $\mathcal{C} = \text{Seq}(\mathcal{A})$ is equivalent to $\mathcal{C}$ is solution of $(\mathcal{C} = \mathcal{E} + \mathcal{A} \times \mathcal{C})$. Then the specification of $\mathcal{C}$ would be this equation joined with the specification of $\mathcal{A}$.

Example 1 (binary words) *We first build the class Σ representing the binary alphabet.*

We can directly say that $\Sigma = \{a, b\}$ with $|a| = |b| = 1$, then we define the specification for $\mathcal{B}$, the class of binary words directly:

$$\mathcal{B} = \text{Seq}(\Sigma)$$

However, another possibility is to use our atom class from earlier. We can consider the class $\Sigma = \mathcal{Z} + \mathcal{Z}$. The resulting class does contain two distinct elements due to our definition for the union.

Then it is possible to use the previous definition, or a recursive specification:

$$\mathcal{B} = \mathcal{E} + \Sigma \times \mathcal{B}$$

Now that we have combinatorial class and the capacity to build them, we can define an important tool to study them.

Definition 8 (ordinary generating function) *Let $\mathcal{C}$ be a combinatorial class, and (C_n) its counting sequence. The Ordinary Generating Function (OGF) of $\mathcal{C}$ is $C(z) = \sum_{n \in \mathbb{N}} z^n C_n$.*

We denote by ρ_C its radius of convergence.

A notation often used to obtain C_n from C is $[z^n]C(z) = C_n$.

The letter used for a combinatorial class will always be the same one used for its corresponding OGF.

Finally, we will often alternate between seeing C as a formal series or a power series. Context can be used to differentiate the two.

Example 2 (binary words, again) For $\mathcal{B}$ the class defined above, we have

$$B_n = 2^n \text{ and } B(z) = \sum_{n \in \mathbb{N}} z^n 2^n = \frac{1}{1-2z}.$$

Property 1 (product, union and sequence on OGF) Let $\mathcal{A}$ and $\mathcal{B}$ be two combinatorial class.

- If $\mathcal{C} = \mathcal{A} \times \mathcal{B}$ then $C(z) = A(z)B(z)$.
- If $\mathcal{C} = \mathcal{A} + \mathcal{B}$ then $C(z) = A(z) + B(z)$.
- If A has no objects of size 0 and $\mathcal{C} = \text{Seq}(\mathcal{A})$, then $C(z) = \frac{1}{1-A(z)}$.

Proof 1 For product, an element of $\mathcal{C}$ of size n has one element of $\mathcal{A}$ of size m and one of $\mathcal{B}$ of size $n - m$. m can have $n + 1$ values. This disjunction on the size of n is basically the Cauchy product which directly appears in the analytic viewpoint.

The union is direct.

For the sequence, write $\mathcal{C} = \mathcal{E} + \mathcal{A} \times \mathcal{C}$. This gives $C(z) = 1 + A(z)C(z)$ which leads directly to $C(z) = \frac{1}{1-A(z)}$.

Now we can take a look at a slightly more complex example:

Example 3 (Rooted binary trees) A specification for the class of binary trees $\mathcal{B}$ is $\mathcal{B} = \mathcal{Z} + \mathcal{Z} \times \mathcal{B} \times \mathcal{B}$. A tree is either a leaf, or a node with two children which are also trees. The size of a tree is for this definition its number of nodes and leaves. If, for instance, we wanted the class of binary trees with only inner nodes counting for sizes, then we would have: $\mathcal{B} = \mathcal{E} + \mathcal{Z} \times \mathcal{B} \times \mathcal{B}$.

We deduce from the combinatorial equation an OGF equation: $B(z) = z + zB^2(z)$ (Here we see B as a formal series). For a fixed real positive z between 0 and $\frac{1}{2}$, this equation admits exactly one positive solution. Therefore, only one OGF is solution of this equation: $B(x) = \frac{1 - \sqrt{1-4x^2}}{2x}$ (Here, B is seen as a power series over real positive numbers). Therefore we have the unicity of $\mathcal{B}$ up to an isomorphism for this problem.

2.2 Boltzmann samplers

This section proposes a sampler on objects of combinatorial class.

We first introduce the boltzmann model: a probability distribution on a combinatorial class.

Definition 9 (Boltzmann models) Let $\mathcal{C}$ a combinatorial class, C its OGF and $x \in]0, \rho_C[$.

The boltzmann model of parameter x over $\mathcal{C}$ assign to each $\gamma \in \mathcal{C}$ the probability:

$$\mathbb{P}_{x,\mathcal{C}}(\gamma) = \frac{x^\gamma}{C(x)}$$

$\mathbb{P}_{x,\mathcal{C}}$ can sometimes be written $\mathbb{P}_x$.

We can easily check that this is a proper definition of a probability over $\mathcal{C}$.

Important to note that it is uniform for a fixed size, i.e. $\mathbb{P}_x(\gamma || \gamma| = n)$ is uniform.

Definition 10 (Boltzmann Samplers) *A boltzmann sampler ΓC for a class $\mathcal{C}$ is an algorithm that produces objects from $\mathcal{C}$ with respect to $\mathbb{P}_x$.*

Then, we note $\Gamma C(x) \in \mathcal{C}$, an output of this algorithm.

This paper will now focus on the boltzmann sampler introduced in [4].

Let $\mathcal{C}$ a specifiable combinatorial class, we now define the Boltzmann Sampler $\Gamma C(x)$ by induction on the specification of $\mathcal{C}$.

- If $\mathcal{C}$ is finite, then this is just taking a random element from C according to a finite distribution which we assume can easily be done.
- If $\mathcal{C} = \mathcal{A} + \mathcal{B}$. We define $p = \frac{A(x)}{A(x)+B(x)} = \frac{A(x)}{C(x)}$. The probability that an element $\gamma \in \mathcal{C}$ is an element of $\mathcal{A}$ is $\mathbb{P}(\gamma \in \mathcal{A}) = p$. Therefore, $\Gamma C(x)$ is $\Gamma A(x)$ with probability p and $\Gamma B(x)$ with probability $1 - p$.
- If $\mathcal{C} = \mathcal{A} \times \mathcal{B}$. Then we observe that $\mathbb{P}_{x,\mathcal{C}} = (\mathbb{P}_{x,\mathcal{A}}, \mathbb{P}_{x,\mathcal{B}})$. Therefore $\Gamma C(x)$ is $(\Gamma A(x), \Gamma B(x))$.
- If $\mathcal{C} = \text{Seq}(\mathcal{A})$. We can again rewrite this equality as: $\mathcal{C} = \{\epsilon\} + \mathcal{A} \times \mathcal{C}$ and use the sampler for product and unions.

We can consider an alternate but equivalent definition of the sequence sampler: because the probability p defined for union is equal to $p = 1 - A(x)$ for $\mathcal{C} + \mathcal{A} \times \mathcal{C}$, we can consider the recursive function:

If Bernoulli of parameter p returns 1, then $\Gamma C(x) = \mathcal{C}$. Otherwise, $\Gamma C(x)$ is equal to $(\Gamma A(x), \Gamma C(x))$ with a recursive call.

Property 2 (correctness and complexity) *For any $x \in]0, \rho_C[$, $\Gamma C(x)$ follows the probability distributions $\mathbb{P}_x$, thus the algorithm is correct.*

Given an oracle computing values of all OGF present in the specification of $\mathcal{C}$ in $O(1)$ the algorithm is linear in the size of its output in terms of number of basic real-arithmetic operations $(+, -, \times, \div)$.

Proof 2 *The construction algorithm gives parse trees in a very similar way to context-free grammar. Then an object $\Gamma C(x)$ has two possible sizes definition: the number of nodes in its parse tree and its size as a combinatorial object. We can establish an important property on those sizes.*

The $\tau(x)$ parse tree associated with an object $\Gamma C(x)$ has a linear size with respect to $|\Gamma C(x)|$.

Let M be the number of combinatorial operations appearing in the specification of $\mathcal{C}$ ($\times$ and $+$, not Seq).

We consider a branch with $M + 1$ elements where each node has the same size n . This branch has two nodes associated with the same class $\mathcal{A}$ appearing

in the specification of $\mathcal{C}$. This is due to the pigeonhole principle. This leads to an infinite number of elements of size $|\tau(x)|$ in $\mathcal{C}$ which is absurd. Therefore such branch can have at most M elements.

A direct consequence of this property is that a subtree of $|\tau(x)|$ with a root of size 0 has at most 2^M elements. (Each node has at most two children).

Therefore, we can deduce by induction on the size of a root a precise majoration: $|\tau(x)| \leq 2^M M |\Gamma C(x)|$.

The algorithm is then linear in the size of $\tau(x)$, indeed $|\tau(x)|$ is the number of calls made to Boltzmann samplers.

2.3 The Rejection Sampler

Our objective, from the start, is to obtain a sampler, uniform for a fixed size, which return objects in a certain size range (and is linear). From this Boltzmann Sampler defined above, we might be able to build a more precise one, but we first need to have a finer understanding on the probability over the size of its output.

Let $N = |\gamma|$, the size of an element of $\mathcal{C}$. Because N is a random variable, we can obtain many properties on its distribution.

Property 3 (probability of N and some notations.) *The probability distribution on N is as follows: $\forall n \in \mathbb{N}, \mathbb{P}_x(N = n) = \frac{C_n x^n}{C(x)}$.*

From this, we can easily deduce the expected value, the second moment and standard deviation:

- $v_1(x) = \mathbb{E}_x(N) = \frac{x C'(x)}{C(x)}.$
- $v_2(x) = \mathbb{E}_x(N^2) = \frac{x^2 C''(x) + x C'(x)}{C(x)}.$
- $\sigma(x) = \sqrt{v_2(x) - v_1(x)^2}.$

Proof 3 *Immediate due to the definition of $\mathbb{P}_x$.*

The formula for the first and second moment are of particular importance. The first determines a certain target size and the second control the Boltzmann Sampler range.

Is is also important to note that $\mathbb{E}_x[N]$ is a strictly increasing function of x as long as $\mathcal{C}$ contains at least two elements of different sizes. Indeed,

$$\mathbb{E}_x[N]' = \frac{\sigma^2(x)}{x}$$

and the standard deviation ($\sigma(x)$ here) is always strictly positive when the associated random variable can take different values.

As we are interested in the asymptotic behaviour of our sampler, we would like to ensure that for each $n \in \mathbb{N}$ we can always choose a x such that $v_1(x_n) = n$. This leads to an important (but not always necessary) condition: the mean value condition $\lim_{x \rightarrow \infty} v_1(x) = \infty$.

We also define a new sampler: the rejection sampler.

Definition 11 (*Rejection Sampler*) For $\mathcal{C}$ a combinatorial class, $n \in \mathbb{N}$, and $\epsilon > 0$ we define the rejection sampler $\mu C(n, \epsilon)$:

Choose $x \in]0, \rho_C[$, sample $\gamma \in \mathcal{C}$ according to $\mathbb{P}_x$, repeat until $|\gamma| \in [n(1 - \epsilon), n(1 + \epsilon)]$. Return γ .

When the mean value condition is satisfied, we will basically always choose x as solution of $v_1(x) = n$. We can still build the rejection sampler when this condition is not satisfied.

As mentionned above, the core idea is to relax the requirement on the size of the output object.

An important aspect to highlight is that assuming the existence of the oracle mentionned earlier for calculating generating functions, the boltzmann sampler and rejection sampler only need the specification of a combinatorial class. If we build such a rejection sampler, we could for both binary words and trees have a working rejection sampler which doesn't require any additional coding expect providing their specification which are simple as seen above. This is also true for any other class built with a specification.

3 The pointing method

The initial promise was to have a linear sampler, this section offer some tools and results related to the rejection sampler complexity. In some cases, we obtain directly a linear complexity. This is not always true, for instance when the mean value condition is not satisfied. When such a situation occurs, one can use the pointing method to solve it. As it will be seen later on, this method can be fully automated.

3.1 sufficient condition for the linearity of Boltzmann Sampler

This section introduces a certain set of combinatorial class for which the Boltzmann Sampler has a linear complexity.

We first introduce the Δ -analytic functions:

Definition 12 (Δ -analytic function) Let f be an analytic function at 0 of radius $\rho > 0$. It is said that f is Δ -analytic if it satisfies the two following conditions:

(i) There exists some $r > \rho$ and some $\theta \in]0, \pi/2[$ such that f can be continued in a domain:

$$\Delta(r, \theta) = \{z | z \neq \rho, |z| < r, \arg(z - \rho) \notin [-\theta, \theta]\}$$

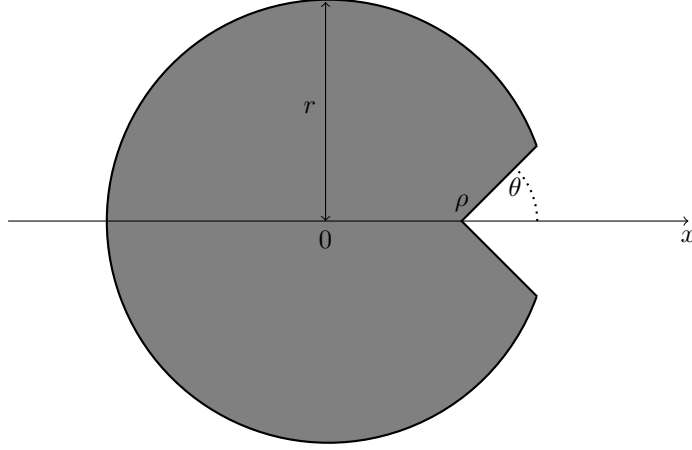
(ii) For z tending to ρ in the domain Δ , $f(z)$ can be approximated as follows:

$$f(z) \sim P(z) + c_0(1 - \frac{z}{\rho})^\alpha + o((1 - \frac{z}{\rho})^\alpha)$$

with $\alpha \in \mathbb{R} \setminus \mathbb{N}$.

The quantity α is called the singular exponent of $f(z)$.

The following drawing helps visualize the domain $\Delta(r, \theta)$:



We now prove a simple transfer theorem which allow the transition from an estimation on a power series, to an asymptotic estimation on its coefficient.

Theorem 1 (Transfer Theorem) *We restrict ourselves to the case α strictly negative.*

Then, for $f(z)$ a Δ -analytic function of singular exponent α ,

$$[z^n]f(z) \sim \frac{c_0 \rho^{-n} n^{-\alpha-1}}{\Gamma(\alpha)}$$

Proof 4 *Under assumption that $\alpha < 0$, the equivalence in (ii) gives: $f(z) = c_0(1 - \frac{z}{\rho})^\alpha + o((1 - \frac{z}{\rho})^\alpha)$. We then have two quantities to estimate.*

Thanks to a theorem proved in the second section of [2] (see corollary 1) we obtain that

$$[z^n]o((1 - \frac{z}{\rho})^\alpha) = o(\rho^{-n} n^{-\alpha-1})$$

By derivating successively the first quantity, we can have an explicit formula for the coefficient of our function:

$$[z^n]c_0(1 - \frac{z}{\rho})^\alpha = c_0 \rho^{-n} \frac{-\alpha(1-\alpha)\dots(n-\alpha-1)}{n!} = c_0 \rho^{-n} \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}$$

The gamma value can be approximated using the Stirling formula [7]:

$$\begin{aligned} [z^n]c_0(1 - \frac{z}{\rho})^\alpha &\sim \frac{c_0 \rho^{-n}}{\Gamma(-\alpha)} \frac{\sqrt{2\pi(n-\alpha+1)}(n-\alpha+1)^{n-\alpha+1} e^{-n+\alpha-1}}{\sqrt{2\pi n n^n e^{-n}}} \\ &\sim c_0 \frac{\rho^{-n} n^{-\alpha-1}}{\Gamma(\alpha)} \end{aligned}$$

Combining both results: we find that

$$[z^n]f(z) \sim \frac{c_0 \rho^{-n} n^{-\alpha-1}}{\Gamma(\alpha)}$$

Starting from this result, [4] proves an important theorem (theorem 6.3 in the paper):

Theorem 2 (Rejection sampler complexity for $\alpha < 0$) *Let $\mathcal{C}$ a combinatorial class with a generating function Δ -analytic of exponent α strictly negative.*

For a fixed $\epsilon > 0$, the rejection sampler $\mu C(n, \epsilon)$ succeeds on average in $O(1)$ for n growing towards infinity where the element x in the rejection sampler is chosen as the unique solution of $n = x \frac{C(x)}{C'(x)} = v_1(x)$ (The mean value condition is satisfied here).

We can take a look at two combinatorial class where this theorem applies.

Example 4 (Compositions, and binary words again) *For binary words, we had seen that $B(z) = \frac{1}{1-2z}$. We deduce that the OGF associated to binary words is Δ -analytic of singular exponent -1 . The rejection sampler for binary words is therefore linear.*

A composition of an integer $n \geq 0$ is an ordered partition of n , i.e. a sequence of strictly positives integers: $(n_1, \dots, n_k)$ such that $n_1 + \dots + n_k = n$. We can then consider the class of compositions of all integers $\mathcal{C}$, an element of size n in $\mathcal{C}$ is a composition of n .

A specification for $\mathcal{C}$ is $\mathcal{C} = \text{Seq}(\mathcal{Z}\text{Seq}(\mathcal{Z}))$. From this, we deduce the associated OGF, $C(z) = \frac{1-z}{1-2z}$ which admits a singularity at $\rho = \frac{1}{2}$. The associated singular exponent is -1 .

3.2 Introducing the pointing method

The goal of this section is to extend the theorem we have just obtained to any singular exponent $\alpha \notin \mathbb{N}$.

We now give a simple theorem which will prove itself very useful.

Theorem 3 (Effect of derivation on a singular exponent) *Let $f(z)$ be a Δ -analytic function of exponent α . Then, $zf'(z)$ is also Δ -analytic of exponent $\alpha - 1$.*

Proof 5 *The first condition (i) is stable by derivation and multiplication.*

For (ii), we only need to prove that the error term is stable by differentiation: if $f(z) = o((1 - \frac{z}{\rho})^\alpha)$ then $f'(z) = o((1 - \frac{z}{\rho})^{\alpha-1})$.

To do so, we will reduce the Δ domain we have to a smaller one: let $\theta' \in]\theta, \frac{\pi}{2}[$ and $r' \in]\rho, r[$. We also consider a small radius k such that for all $z \in \Delta(r', \theta')$, the circle C of radius $k|z - \rho|$ and centered on z verifies $C \subset \Delta(r, \theta)$. We note $\gamma(z)$ its contour.

Let $z \in \Delta(r', \theta')$. By Cauchy formula, we have $g'(z) = \int_{\gamma(z)} \frac{g(w)dw}{(z-w)^2}$. We observe three points: the contour $\gamma(z)$ is of length $2\pi k|z - \rho|$; for $w \in \gamma(z)$, $|z - w| = k|z - \rho|$; and by some ϵ operations, we obtain that $\int_{\gamma(z)} |g(w)|d(w) = o((\rho - z)^\alpha)$. All three elements combined give

$$g'(z) = o\left(\left(1 - \frac{z}{\rho}\right)^{\alpha-1}\right)$$

The proof was taken from [6], page 419.

This theorem shows that derivating a function allows to modify the value of α , this is the main idea behind the pointing method.

To use it in our combinatorial class model, we need to consider an operation corresponding somewhat to the derivation:

Definition 13 (The pointing operator) Let $\mathcal{C}$ a combinatorial class.

The pointed class of $\mathcal{C}$ is $\mathcal{C}^\bullet = \{(c, C), c \text{ is an atom of } C, C \in \mathcal{C} \text{ s.t. } |(c, C)|_{\mathcal{C}^\bullet} = |C|_{\mathcal{C}}\}$.

The atoms of C where mentionned and explained just after the definition of a specification.

Since $C_n^\bullet = nC_n$, the pointing operator corresponds to the derivation of the OGF (and multiplication by the identity).

We note by $\mathcal{C}^n$ the n-th iteration of the pointing operator.

Theorem 4 (Pointing operator and specifiable class) Let $\mathcal{C}$ be a specifiable combinatorial class. Then $\mathcal{C}^\bullet$ is also specifiable.

This theorem justifies speaking of the Boltzmann Sampler $\Gamma C^\bullet$ of a pointed class $\mathcal{C}^\bullet$ as long as $\mathcal{C}$ is specifiable. From $\Gamma C^\bullet(x)$, it is easy to obtain the associated element of $\mathcal{C}$. The resulting probability distribution on $\mathcal{C}$ is still uniform under hypothesis of fixed size.

Proof 6 The core idea behind the specification of $\mathcal{C}^\bullet$ is that for each auxiliary class $\mathcal{F}$, we can also consider a new auxiliary class $\mathcal{F}^\bullet$.

All three operators seen above have composition rules with the pointing operator:

$$\begin{aligned} (\mathcal{A} + \mathcal{B})^\bullet &= \mathcal{A}^\bullet + \mathcal{B}^\bullet \\ (\mathcal{A} \times \mathcal{B})^\bullet &= \mathcal{A}^\bullet \times \mathcal{B} + \mathcal{A} \times \mathcal{B}^\bullet \\ (\text{Seq}(\mathcal{A}))^\bullet &= \text{Seq}(\mathcal{A}) \times \mathcal{A}^\bullet \times \text{Seq}(\mathcal{A}) \end{aligned}$$

For any equation of combinatorial class: $\mathcal{F} = \Psi(\mathcal{F}_1, \dots, \mathcal{F}_i)$, there exist by induction an equation for the pointed class $\mathcal{F}^\bullet = \psi(\mathcal{F}_1, \dots, \mathcal{F}_i, \mathcal{F}_1^\bullet, \dots, \mathcal{F}_i^\bullet)$.

Then the system of equations for the specification of $\mathcal{C}$ can be turned into a specification of $\mathcal{C}^\bullet$ by adding for each equation of an auxiliary class, the associated pointed equation.

Thanks to theorems 3 and 4, we can deduce an extension of the theorem 2:

Theorem 5 (Rejection sampler complexity for $\alpha \notin \mathbb{N}$) *We define a new rejection sampler: $\mu C^k(n, \epsilon)$. This rejection sampler simply points k times the class $\mathcal{C}$, then calls the rejection sampler defined earlier. The result is then an element of $\mathcal{C}$ with k pointed elements which can be removed.*

Let $\mathcal{C}$ a combinatorial class with a generating function Δ -analytic of exponent $\alpha \notin \mathbb{N}$. There exists a $k \in \mathbb{N}$ such that, for all fixed $\epsilon > 0$, the rejection sampler $\mu C^k(n, \epsilon)$ succeeds on average in $O(1)$ for n growing towards infinity.

Proof 7 *We can simply choose k such that the class $\mathcal{C}^k$ is of singular exponent negative. Then we use the rejection sampler defined above which choose x using the mean value condition.*

3.3 Closure properties on Δ -analytic functions and combinatorial class

All theorems seen in this section have used as a starting condition that the OGF of some class $\mathcal{C}$ was Δ -analytic. A core element of Boltzmann Samplers is the automation of its process, but this can't be done if we need to study for each class the OGF to see if it is Δ -analytic. Sadly, there is no general result to obtain that all specifiable class are of this family. All we can do is to obtain for some smaller families this result.

This section will present two of those sets.

We start by the family for non recursive specifications:

Theorem 6 (Closure properties on non recursive specification) *Let $\mathcal{A}$ and $\mathcal{F}_1, \dots, \mathcal{F}_n$ be combinatorial class. Let Ψ be a formula on combinatorial class using only unions, products and sequences. If $\mathcal{A} = \Psi(\mathcal{F}_1, \dots, \mathcal{F}_n)$ and all $\mathcal{F}_i$ have a Δ -analytic OGF, then $\mathcal{A}$ also has a Δ -analytic OGF.*

Proof 8 *By induction, we only need to establish this theorem for three special cases:*

- $\Psi(\mathcal{A}, \mathcal{B}) = A \times B$
- $\Psi(\mathcal{A}, \mathcal{B}) = A + B$
- $\Psi(\mathcal{A}) = \text{Seq}(A)$

For addition and multiplication, if A and B have different singularities, then we can ignore the higher one. Therefore, we can assume they both have the same singularity ρ .

Let $A(z) = P(z) + a(1 - \frac{z}{\rho})^\alpha + o((1 - \frac{z}{\rho})^\alpha)$ and $B(z) = Q(z) + b(1 - \frac{z}{\rho})^\beta + o((1 - \frac{z}{\rho})^\beta)$. Without loss of generality, we assume that $\alpha \leq \beta$.

If $\alpha < \beta$, then we can neglect most of $B(z)$, $B(z) - Q(z) = o((1 - \frac{z}{\rho})^\alpha)$. Then,

$$A(z) + B(z) = Q(z) + P(z) + a(1 - \frac{z}{\rho})^\alpha + o((1 - \frac{z}{\rho})^\alpha)$$

Otherwise, $A(z)$ and $B(z)$ both have their contribution:

$$A(z) + B(z) = Q(z) + P(z) + (a + b)\left(1 - \frac{z}{\rho}\right)^\alpha + o\left(\left(1 - \frac{z}{\rho}\right)^\alpha\right)$$

For multiplication,

$$A(z)B(z) = P(z)Q(z) + c\left(1 - \frac{z}{\rho}\right)^\delta + o\left(\left(1 - \frac{z}{\rho}\right)^\delta\right)$$

Where c and δ depend on β . If $\beta \leq 0$, then $c = ab$ and $\delta = \alpha + \beta$. Otherwise, $c = aQ(\rho)$ and $\delta = \alpha$. It is important to note that in both cases, δ is not a natural integer.

Last case is the sequence, we just need to prove that $\frac{1}{1-A(z)}$ is Δ -analytic. This is a special case of composition which has been covered in [6], see section 6.9 for more informations.

Surprisingly, another family of this type is in complete opposition to the one we just studied.

Theorem 7 (Closure properties on recursive specifications) *Let $\mathcal{C}$ be a specifiable combinatorial class. We say that an auxiliary class $\mathcal{A}$ depends on another $\mathcal{B}$ if $\mathcal{B}$ appears in the equation defining $\mathcal{A}$. From this we deduce an oriented graph of dependance on the specification of $\mathcal{C}$.*

We also introduce $d = \text{pgcd}(\{n | C_n \neq 0\})$ which we will call the periodicity of $\mathcal{C}$.

If $\mathcal{C}$ admits a specification which has a strongly connected graph of dependance and $\mathcal{C}$ is of periodicity $d = 1$, then $C(z)$ is Δ -analytic of singular exponent $\frac{1}{2}$.

This result can be tricky to demonstrate and we let the reader look at [4] and [6], section 7.6 to obtain the proof.

4 Conclusion

Our initial promise was to have a linear sampler on combinatorial structures when aiming at a certain range of sizes which is uniform for a fixed size.

To do so, we first defined combinatorial class and a probability distribution on those structures: the Boltzmann model. We easily built a linear sampler on this distribution and deduced a restriction to a certain range of sizes with the Rejection Sampler. The last remaining question was on the complexity of this rejection sampler as it would require on average a constant number of calls to the Boltzmann Sampler to ensure linearity. To answer this question, we looked at Δ -analyticity and established that class with an ordinary generating function Δ -analytic in almost all cases are valid for the rejection sampler. To obtain this last result, we used the pointing method and therefore justified its utility (which was the first aim of this paper). Finally we saw some limited closure properties on Δ -analytic functions.

To make this paper readable, some part of the literature on the subject were ommited. There are two appendix. In the first we offer to establish more advanced operations for combinatorial class. For the second, we look at the possibility of considering a sampler at the singularity, i.e. taking $x = \rho_C$ by slightly modifying the definition of a boltzmann sampler. Another restriction made in this paper was to only look at unlabelled structures, labelled structures use exponential generating function (in opposition to ordinaty). Despite this change both have very similar results. Sadly, due to a lack of place, labelled structures didn't make it to an appendix. Finally, despite earlier claim that there are no general closures properties on Δ -analyticity, i.e. all specifiable class are of this family, the theory on algebraic function maybe offers such a property (the author of this paper insist on the maybe). This topic can be further explored in [6].

Appendix A: More combinatorial class

For this paper, we have only studied basic combinatorial operations: the product, union and sequence. We now introduce three more complex constructions taken from [5].

Definition 14 (Set) *Let $\mathcal{C}$ be a combinatorial class, the set of $\mathcal{C}$ is defined as $Set(\mathcal{C}) := \{I | I \subset \mathcal{C}, \text{ and } I \text{ is finite}\}$. Then the size of I is $|I| = \sum_{i \in I} |i|$ (not to be confused with the cardinality of I).*

Definition 15 (Multiset) *By definition, the set construction doesn't allow repetition. If we want them, we can use the multiset construction: $M(\mathcal{C})$ is the class of collections of elements of $\mathcal{C}$ with repetitions allowed. A more formal definition could be done in a similar way to the union, using tokens of size 0 to differentiate same elements of $\mathcal{C}$.*

Definition 16 (Cycle) *The cycle construction is defined as*

$$Cycle(\mathcal{C}) := \{(a_0, \dots, a_k) | k \in \mathbb{N}, a_0, \dots, a_k \in \mathcal{C}, a_0 = a_k\}$$

. *Moreover, we consider that two elements are equal if and only if they are a cyclic shift of each other, therefore elements are not ordered. Like above, the size of a cycle is taken as the sum of its elements.*

Those construction admit Boltzmann Sampler like union, product and sequences. However, they are far more complex. They also have formulas associated with their generating function.

Most theorems seen above are also valid for sets, multisets and cycles.

Example 5 (Partitions of integer) *We have defined above, the class of compositions. The class of partition is very similar: a partition of an integer $n \geq 0$ is a multiset of strictly positives integers: $\{n_1, \dots, n_k\}$ such that $n_1 + \dots + n_k = n$.*

We can then consider the class of partitions of all integers $\mathcal{P}$, an element of size n in $\mathcal{P}$ is a partition of n .

We can define $\mathcal{P}$ using tools defined above: $\mathcal{P} = \text{MSET}(\mathcal{Z} \times \text{Seq}(\mathcal{Z}))$. The same can be done while only considering partitions with non repeting integers: $\mathcal{Q} = \text{MSET}(\mathcal{Z} \times \text{Seq}(\mathcal{Z}))$.

As repeated earlier, by only using these specifications we have a working sampler.

Appendix B: Boltzmann Sampler at the singularity

We have restricted ourselves to the study of Boltzmann Sampler with a parameter x being stricly inferior to the singularity ρ . What happens when we consider $x = \rho$?

When $C(\rho) < \infty$, since $C'(\rho) \neq 0$ we have $v_1(\rho) < \infty$ which as explained above gives poor complexity.

So we restrict ourselves to $C(\rho) = \infty$, this has one major pro and one major con: we can calculate once and for all $A(x)$ for each auxiliary class $\mathcal{A}$, but we need to modify our samplers to work with the infinity. In [4], the author offer a broad approach to this problem, but here we will only study one example from the same paper.

Example 6 (Singular sampler for sequences) *Let $\mathcal{A}$ and $\mathcal{C}$ be two combinatorial class such that: $\mathcal{C} = \text{Seq}(\mathcal{A})$ and $\rho_C < \rho_A$. This way we can still consider $A(\rho_C)$. Then we adapt our boltzmann sampler on sequence.*

The recursive definition used a Bernouilli law with $p = 1 - A(x)$ as parameter. This probabity verifies at the singularity, $p \leq 0$, therefore the sampler never ends.

Instead, we can stop the sampler when the size of all elements combined in the sequence have reached the goal.

Under minors assumptions on $\mathcal{A}$, this new sequence sampler offers two advantages: the time complexity is still linear and the expected size output is $n + O(1)$, which means that in opposition to the previous samplers, this sampler is more and more precise as n grows. The paper [4] offers more precise informations.

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